

# SOFTWARE ENGINEERING LAB 1

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## Section E

### 1. Requirements Table

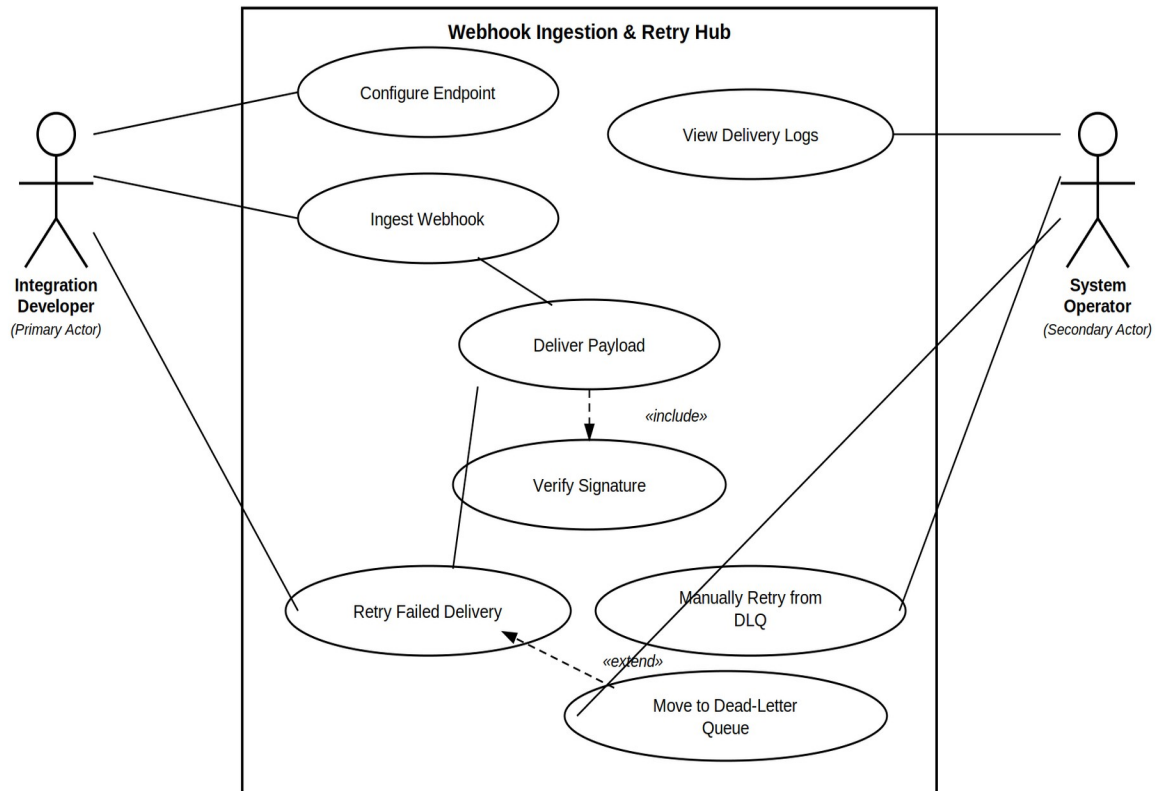
5 Functional Requirements (FR-001 – FR-005) and 2 Non-Functional Requirements (NFR-001 – NFR-002).

| ID      | Type                   | Description                                                                                                                                                                 | Priority | Acceptance Criteria                                                                                                                                  | Rationale                                                                                                          |
|---------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| FR-001  | Functional             | The system shall ingest inbound webhooks, forward payloads to target endpoints, and automatically retry failed deliveries using exponential backoff (1s, 2s, 4s, 8s).       | High     | Pass: Failed delivery successfully retried and logged. Fail: Dropped webhook without dead-letter persistence.                                        | Guarantees no event is silently lost during transient downstream failures.                                         |
| FR-002  | Functional             | The system shall persist the raw payload, headers, source IP, and timestamp of every inbound webhook before attempting delivery.                                            | High     | Pass: Every ingested webhook has a corresponding stored record queryable by ID. Fail: Payload not retrievable after ingestion.                       | Provides an audit trail and enables replay/debugging of past events.                                               |
| FR-003  | Functional             | The system shall allow a System Operator to manually trigger a retry for any webhook currently in the dead-letter queue.                                                    | Medium   | Pass: Manually triggered event is redelivered and its status updates to Success/Failed. Fail: Manual trigger has no effect on delivery state.        | Operators need a manual recovery path for events that exhausted automatic retries.                                 |
| FR-004  | Functional             | The system shall allow an Integration Developer to register, update, and deactivate target delivery endpoints, including a signing secret for each.                         | High     | Pass: New/updated endpoint configuration is used on the next delivery attempt. Fail: Deliveries continue to use stale endpoint configuration.        | Developers must be able to onboard and maintain integrations without code deployment.                              |
| FR-005  | Functional             | The system shall expose the delivery status (Pending, Delivered, Retrying, Dead-Lettered) of each webhook event via a dashboard or API.                                     | Medium   | Pass: Status returned matches the actual internal delivery state at query time. Fail: Status is stale or unavailable.                                | Gives both actors visibility into pipeline health without inspecting raw logs.                                     |
| NFR-001 | Performance & Security | The ingestion endpoint shall buffer inbound webhooks with 99.999% reliability and throughput up to 2,500 requests/second.                                                   | High     | Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.                                                    | The hub must not become a bottleneck or single point of failure for upstream partners sending high-volume traffic. |
| NFR-002 | Security               | The system shall verify the HMAC signature of every inbound webhook against the registered endpoint secret before it is queued for delivery, and reject unsigned or invalid | High     | Pass: Requests with invalid/missing signatures are rejected with an audit log entry and never reach downstream services. Fail: An unsigned or forged | Prevents spoofed or tampered events from                                                                           |

| ID | Type | Description | Priority | Acceptance Criteria   | Rationale                                          |
|----|------|-------------|----------|-----------------------|----------------------------------------------------|
|    |      | requests.   |          | payload is delivered. | being trusted and forwarded into internal systems. |

## 2. UML Use-Case Diagram

Webhook Ingestion & Retry Mechanism Hub - Use Case Diagram



### 3. Use-Case Flow Specification

#### Use Case: Ingest and Deliver Webhook (with Retry)

**Actors:**

- Primary: Integration Developer's external system (as the triggering third party)
- Secondary: System Operator (notified on dead-letter events)

**Preconditions:**

- The target endpoint has been registered and configured with a valid signing secret (FR-004).
- The ingestion service is running and reachable.

**Postconditions:**

- Success: The webhook payload has been delivered to the target endpoint and its status is recorded as Delivered.
- Failure: All retry attempts are exhausted and the event is persisted in the dead-letter queue with status Dead-Lettered.

**Main Success Scenario:**

1. The external system sends an HTTP POST webhook request to the ingestion endpoint.
2. The system verifies the HMAC signature of the request against the registered secret («include» Verify Signature).
3. The system persists the raw payload, headers, and timestamp, and assigns the event status Pending (FR-002).
4. The system forwards the payload to the configured target endpoint.
5. The target endpoint responds with a success status code (2xx).
6. The system updates the event status to Delivered and logs the successful delivery.

**Alternate Flow — Delivery Failure with Retry:**

7. At step 5, the target endpoint fails to respond or returns a non-2xx status code.
8. The system marks the event status as Retrying and schedules a retry using exponential backoff (1s, 2s, 4s, 8s).
9. The system re-attempts delivery at each backoff interval, repeating steps 4–5.
10. If a retry succeeds, the flow rejoins the Main Success Scenario at step 6.
11. If all retry attempts are exhausted without success, the system triggers the «extend» use case Move to Dead-Letter Queue: the event status becomes Dead-Lettered and the System Operator is notified for manual review (FR-003).